CSC 4520/6520: Design & Analysis of Algorithms Spring 2026

Instructor: Prakash Chourasia

Spring 2026

Course Information

Hours:	MW 10:00 am – 11:40 am
Location:	Classroom South 101
Office:	1 Park Place, Office 956 (DICE and Bioinformatics Center)
E-mail:	pchourasia1@gsu.edu (NO iCollege)
Website:	My webpage and Here you can read my research work
Office Hours:	MW 1:00 pm – 2:00 pm or by appointment

Teaching Assistant

TA:	Yixian Chen
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Office:	25 Park Place, 6th Floor INSPIRE Lab
Office Hours:	MW 2:00 pm – 3:00 pm or by appointment

1 Course Overview

This course will teach you how to analyze problems and solve them using computational procedures called algorithms. We will cover many algorithms foundational to Computer Science and learn their uses, advantages, and disadvantages. Most importantly, we will develop skills and intuition to help you design efficient algorithms for new problems.

Learning Outcomes

By the end of this course, you will be able to:

- Use brute force, decrease-and-conquer, divide-and-conquer, transform-and-conquer, greedy programming, and dynamic programming techniques to solve problems.
- Implement and employ a variety of advanced data structures to solve problems, including graphs, self-balancing trees, and trees.
- Implement a variety of algorithms to sort, search, and traverse data.
- Determine the time and space efficiency of algorithms using Big O, Big Theta, and Big Omega notation in terms of common complexity classes.

- Identify the appropriate use cases of these algorithms, data structures, and programming techniques, and evaluate the tradeoffs of possible solutions, including time and space complexity, to decide on an approach to a problem. The content and skills in this course have a wide variety of applications in both academia and industry. Pragmatically, many tech companies interview candidates around these topics. At the same time, this course provides a mental framework that lets you solve hard problems–the problem-solving techniques taught here show up in various subfields in Math, Engineering, and Computer Science.

Prerequisites & Background

You should have some knowledge of most of the following areas (not limited to):

- Programming in at least one programming language. Preferably in Python 3, since all examples in lectures and all homeworks will be given with the expectation of some knowledge of Python 3.
- You are expected to have taken CSC 2510 Theoretical Foundations of Computer Science and CSC 2720 Data Structures, and received a grade of C or higher in both classes

Textbook

Course materials should be sufficient to do well in this class. The textbooks are optional. While it may be useful for perspective and context into most of our topics, it will not cover the exact same set of topics we'll traverse. If you need a book, I suggest the following:

- "Data Structure and Algorithmic Thinking with Python" by Narasimha Karumanchi, ISBN-13 9788192107592, ISBN-10 8192107590
- Introduction to Algorithms by Cormen, Leiserson, Rivest, and Stein (Fourth Edition, MIT Press, 2022) ISBN: 9780262046305
- Problem Solving with Algorithms and Data Structures Using Python by Miller, and Ranum (Second Edition, Franklin, Beedle & Associates, 2011) ISBN: 9781590282571

2 Tentative Course Plan

The course will cover:

- Data structures, sorting, compression (sorting, hashing, priority queues, binary search trees, coding)
- General algorithmic methods (divide and conquer, greedy algorithms, dynamic programming)
- Combinatorial problems (Eulerian cycle, BFS, DFS, MST, Shortest Path, PageRank, Matchings, Network Flows, and Minimal Cuts)
- Computational Complexity (NP-completeness, reducibility)
- Heuristics and approximation algorithms (Bin packing, TSP, branch-and-bound, local search, simulated annealing, genetic algorithms, Maximum cut, coloring, independent sets)

Date	Topic
Jan 12	Course overview; algorithm analysis basics
Jan 14	Review of fundamental data structures
Jan 19	*University Holiday (MLK Day)*
Jan 21	Priority queues, Stacks
Jan 26	Sorting algorithms: comparison-based
Jan 28	Sorting: non-comparison-based; Linear Algo
Feb 2	Divide-and-conquer algorithms
Feb 4	Recurrence relations and analysis
Feb 9	Solving Recurrence relations and Master Theorem
Feb 10	Greedy algorithms: design principles
Feb 16	*Class Cancelled*
Feb 18	Dynamic programming: fundamentals II
Feb 23	Dynamic programming : Problems
Feb 25	Trees, and Intro to graphs
Mar 2	**Midterm Exam 1**
Mar 4	Intro to Tree
Mar 9	Binary Tree and Heap Sort
Mar 11	Graph represeantation and DFS
Mar 16	*Spring Break — No Class*
Mar 18	*Spring Break — No Class*
Mar 23	Graph algorithms: BFS and DFS
Mar 25	Graph algorithms: Relaxation Algorithm and Kahn's Algorithm
Mar 30	Prims, Kruskals, Dijkstra
Apr 1	A*, Bellman Ford Algorithms
Apr 6	Floyd Warshall, Johnson's Algo
Apr 8	NP-Hard and NP-complete problems
Apr 13	**Midterm Exam 2**
Apr 15	Graph Theory & Randomized Algorithms
Apr 20	Network flow algorithms, Branch-and-bound; local search
Apr 22	Simulated annealing; genetic algorithms; Maximum cut, coloring, independent sets; course review
Apr 27	**Final Exam**
Apr 29	**Class Review - Open floor**

3 Grading Policy

3.1 Homework/Assignments

Assignments permit you to practice and apply what we are learning in class. All homework should be uploaded on iCOLLEGE drop box into the corresponding folders. Submissions should be either MS Word/OpenOffice or pdf documents, as other formats might not be supported for reading. You must follow all requirements for the homework submissions. Ususally each submission should consist of

1. (if required) the source code as a py-file;
2. (if required) the report in a pdf- (preferable) or MS Office format.

The submissions must be uploaded on iCollege dropbox into the relevant folder. If the teaching assistant cannot open and read it, or if you submit a flawed document, the grade will be zero. It is your responsibility to ensure that each file opens correctly. Having submitted it on iCollege, go and try to open/download it. Always make sure that you hit the "submit" button, and get a notification from iCollege that your submission has successfully gone through.

Remark. ***Assignments should be HANDWRITTEN and SCANNED; Unless otherwise specified in the assignment instructions.***

Late submission penalty is a 20 points deduction for each extra 24 hours after the deadline. E.g., a homework scored 83% but submitted one day late (more than 1 hour and less than 24 hours late) will only receive 63% for the grade; the same homework submitted 2 days late (more than 24 hours and less than 48 hours late) will only receive 43%, and so on. Thus, even a perfect 100%-score submission which is more that five-day-late will receive no credit (but may receive some feedback).

Note: DO NOT upload individual images. Please, make a MS Word/OpenOffice or PDF file out of the scanned pages.

3.2 Tests/Exams

Examinations will be conducted during scheduled class periods and are generally announced at least one week in advance. Given the cumulative structure of the course, exams may include material from all topics covered to date. Students are responsible for monitoring iCollege Announcements for any updates.

Remark. *Final Exam: The final exam covers all material covered in the class.*

No make up exams request will be entertained. Exceptions to this policy will be made only in case of a family or medical emergency, which must be clearly documented.

3.3 Quiz

Quizzes will be administered in class as unannounced (surprise) quizzes and may occur at any point during the lecture period. Completion of an in-class quiz will also serve as a record of attendance for that day. No makeup quizzes will be offered.

Remark. ***Quiz score adjustments will be considered only in cases of officially documented absences submitted through the Professor Absence Notification system with the Dean of Students.***

Please note that if you are missing a class or coming late or leaving early, you are risking your quiz scores. Exceptions to this policy will be made only in case of a family or medical emergency, which must be clearly documented.

3.4 Attendance

Attendance of the lectures is highly recommended since this class is one of the most important classes for a future computer scientist and a programmer. In particular we will be discussing and solving a lot of problems that often are asked during technical job interviews. Also, if you are absent, it is your responsibility to find out what you missed and to come to the next class prepared.

Note: Complete the Professor Absence Notification with the Dean of Students at this link Professor Notification of Absence if you experience an emergency that forces you to miss multiple classes.

3.5 Course Grade

Your final course grade will be computed as follows:

Assessment Component	**Percentage**
Homework/Assignmnets (6-7 HW)	30%
Tests (Midterm 1 & 2)	30%
Final Exam	25%
Quizes (5-7 Quizes)	15%
Total	**100%**

Table 1: Grading Breakdown

The scheme could be changed at the instructor's discretion, and you will be notified in advance.

The TA is responsible for grading all assignments, quizzes and exams. While the instructor might provide the grader with guidelines (for example, that a question is worth 10 points), it is the TA's job to determine correctness of your answer. The TA is authorized to adjust grades if there has been a mistake in the grading (for example, if the grades were added incorrectly). Therefore, if you have questions regarding how your work was graded (such as "why were 3 points taken off for this problem?"), please contact the TA. Any questions regarding graded work must be brought to the TA's attention within 2 weeks of the date on which the work was first returned to the class (or grades were reported on iCollege).

4 Academic Integrity

The course is designed so that you do well when you put in the effort. We will make every effort to help you succeed. Should you find yourself in a position where you feel trapped or encouraged to cheat, we hope you will email us instead to ask for help. That being said, here are the expectations around academic honesty.

Cheating

Do not cheat! You can discuss problems with other people, but when you start coding, do it on your own! If you are stuck, you can google how to approach the problem (e.g., stackoverflow.com is a good source of the solutions), however, do not bluntly copy a solution, understand it, then close it and do it yourself with best understanding. For homework, copying solutions from any source (students, websites, textbooks), whether wholesale, in part, or modifying the code afterwards, is strictly considered cheating. Providing solutions to other students or third-party websites, in part or in whole, is also considered cheating.

Copying information from another student's work will be punished severely resulting in the grade zero for this type of activity (homework/exam) for all students involved. In some cases of plagiarism (at the instructor's discretion), students may receive a failing grade for the course.

Collaboration

Note that the above policy does not mean that you have to work alone the entire semester. For homework problems, you're allowed to discuss approaches, provide hints, and key insights with up to five classmates, but must write up solutions on your own. Please cite any students, web sources, etc. that you consulted while solving problems. You should never possess solutions to exact homework questions other than those solutions you have written yourself.

5 Cell Phone Policy

Please turn off your cell phones or put them on silent operation before entering the class. Make sure to take good notes to succeed in this class.

6 Students with Special Needs

Students who need special accommodations should arrange a meeting with the instructor as soon as possible. Bring a copy of your Student Accommodation Form to the meeting. If you do not have this form, make an appointment with the university's Office of Disability Services, 230 New Student Center, 404-413-1560.

This syllabus is intended as a general guide for the course. The instructor reserves the right to make changes as deemed necessary.